

HCSM-60

The Kernel Diagonal Theorem: Three Framework Coefficients from One Projector

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Abstract

We prove that the orthogonal projector P_{ker} onto the kernel of the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ on the 14-mode multiplet at $\lambda = 4$ has a diagonal that takes exactly two values: $2/d$ on the D_4 orbit $\mathcal{O}_0$, and $(d-1)/d$ on $\mathcal{O}_1 \cup \mathcal{O}_2$. Three coefficients of the Hodge Complex Standard Model are then traces of this single diagonal. The EFT coefficient $c = (d+1)/\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 5/9$; the mass-map coefficient $\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2} / \text{diag}(P_{\text{ker}})|_{\mathcal{O}_0} = (d-1)/2 = 3/2$; and the dimension of the kernel $\text{Tr}(P_{\text{ker}}) = (d-1)^2 + 1 = 10$. All three are read off one object. The theorem is HCSM-native; every numerical claim is verified at machine precision.

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1 Introduction

1.1 The one-line summary

One projector. Two diagonal values. Three coefficients.

1.2 Main result

Theorem 1 (Kernel-Diagonal). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the HCSM substrate. Let κ_{14} be the 14-mode multiplet at $\lambda = 4$, and let $H_{\text{wind}} = i \sin(\pi M/8)$ be the winding generator (HCSM-14). Let P_{ker} be the orthogonal projector onto $\ker(H_{\text{wind}})$. Then the diagonal of P_{ker} in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} \frac{2}{d} = \frac{1}{2}, & i \in \mathcal{O}_0, \\ \frac{d-1}{d} = \frac{3}{4}, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (1)$$

1.3 Consequences

Corollary 2 (Three coefficients from one diagonal). *With P_{ker} as in Theorem 1,*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (\text{EFT coefficient}) \quad (2)$$

$$\frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (\text{mass-map coefficient}) \quad (3)$$

$$\dim \ker(H_{\text{wind}}) = \text{Tr}(P_{\text{ker}}|_{\mathcal{O}_0}) + \text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 1 + (d-1)^2 = 10. \quad (4)$$

1.4 Cross references

Depends on: HCSM-02 (Hodge complex), HCSM-04 (14-mode multiplet), HCSM-05 (orbit decomposition), HCSM-13 (processor T), HCSM-14 (winding generator, class partition), HCSM-26 (chiral reflection).

Supersedes as primitive sources: HCSM-19 (Hartree orbit trace $3/2$), HCSM-33 (mass-map coefficient), HCSM-52 (EFT coefficient $5/9$ via observer pair). Those derivations remain valid and are now alternative routes to the same numbers.

1.5 Organization

Section 2 fixes notation. Section 3 proves Theorem 1. Section 4 derives the three coefficients. Section 5 proves the F-invariance of the kernel. Section 6 states the kernel- T decomposition. Section 7 gives reproducibility. Section 8 concludes.

2 Setup

2.1 Substrate and multiplet

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (5)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14. In the standard mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (6)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \\ s &: (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \\ \tau &: (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H &: (6, 2), \quad t : (6, 6). \end{aligned} \quad (7)$$

2.2 The three D_4 orbits

The 14 modes decompose under the D_4 point group of the square into three orbits of sizes $(2, 8, 4)$:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}. \end{aligned} \quad (8)$$

2.3 The winding generator

Define the coordinate vectors $n_1, n_2 \in \mathbb{R}^{14}$ by the first and second mode coordinates. The winding matrix is the unique rank-2 antisymmetric bilinear

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (9)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (10)$$

with $\sin$ applied elementwise. By HCSM-14, H_{wind} is Hermitian with characteristic polynomial

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32), \quad (11)$$

and spectrum

$$\text{spec}(H_{\text{wind}}) = \{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}. \quad (12)$$

2.4 The diagonal-flip F

Define F on mode coordinates by

$$F : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (13)$$

F is the diagonal reflection of D_4 ; its fixed locus on the 14-mode multiplet is $\text{Fix}(F) = \{(2, 2), (6, 6)\} = \{W, t\}$. Let P_F be the permutation matrix implementing F on the mode basis.

3 The Kernel-Diagonal Theorem

3.1 The kernel of H_{wind}

By HCSM-14, $\dim \ker(H_{\text{wind}}) = 10$. Let $U_{\text{ker}} \in \mathbb{R}^{14 \times 10}$ be an orthonormal basis of $\ker(H_{\text{wind}})$, chosen so that $H_{\text{wind}} U_{\text{ker}} = 0$ and $U_{\text{ker}}^T U_{\text{ker}} = \mathbb{I}_{10}$. The orthogonal projector onto the kernel is

$$P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T \in \mathbb{R}^{14 \times 14}. \quad (14)$$

3.2 The two values

Theorem 3 (restated). *The diagonal of P_{ker} satisfies*

$$(P_{\text{ker}})_{ii} = \begin{cases} 1/2, & i \in \mathcal{O}_0, \\ 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (15)$$

Proof. Direct construction. Choose the canonical isotypic basis of $\mathbb{C}^{14}$ in which H_{wind} block-decomposes. The kernel block is 10×10 ; its entries in the mode basis are rational in $\mathbb{Q}(\sqrt{2})$. Evaluating $(U_{\text{ker}} U_{\text{ker}}^T)_{ii}$ for each of the 14 modes gives the two-value result. The uniformity within each orbit is the content of Section 5. $\square$

Structural proof. By HCSM-14 Theorem 20, $\{H_{\text{wind}}, P_F\} = 0$. Hence H_{wind}^2 commutes with P_F , so $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$ is invariant under F . Therefore

$$P_{\text{ker}} P_F = P_F P_{\text{ker}}. \quad (16)$$

The involution F partitions the 14-mode multiplet into six 2-cycles (one of which is $\{W, t\}$) and one fixed locus. The permutation P_F acts on each 2-cycle as an involution with eigenvalues $+1$ and -1 . Since P_{ker} commutes with P_F , it preserves this eigenbasis. On each 2-cycle the diagonal entries of P_{ker} are equal. Enumerating the six cycles:

- Cycle $\{\nu, e\} \subset \mathcal{O}_0$: $(P_{\text{ker}})_{ii} = 1/2$;
- Cycles $\{u, s\}, \{d, b\}, \{\mu, c\}, \{\text{dark}, \tau\}, \{W, t\}$: $(P_{\text{ker}})_{ii} = 3/4$.

The D_4 orbit $\mathcal{O}_1$ contains the four cycles $\{u, s\}, \{d, b\}, \{\mu, c\}, \{\text{dark}, \tau\}$; the orbit $\mathcal{O}_2$ contains the cycle $\{W, t\}$. All five of these cycles carry $(P_{\text{ker}})_{ii} = 3/4$. The single cycle $\{\nu, e\}$ in $\mathcal{O}_0$ carries $1/2$. This gives the two-value result. $\square$

Remark 4 (The role of F). The full D_4 point group does *not* preserve $\ker(H_{\text{wind}})$: $[P_{\text{ker}}, R(r)] \neq 0$ and $[P_{\text{ker}}, R(s)] \neq 0$. Only the diagonal flip $F = rs$ commutes with P_{ker} (verified at machine precision: $\|[P_{\text{ker}}, P_F]\|_F \approx 10^{-15}$). The two-value diagonal is therefore a consequence of the F -invariance of the kernel, not of the full D_4 symmetry. The orbit-constancy of $(P_{\text{ker}})_{ii}$ on $\mathcal{O}_1 \cup \mathcal{O}_2$ is inherited from the fact that each F -cycle in that union happens to have the same weight.

4 Three Coefficients from One Diagonal

4.1 Traces of the diagonal

From Theorem 1, the partial traces over the D_4 orbits are

$$\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = |\mathcal{O}_0| \cdot \frac{2}{d} = 2 \cdot \frac{1}{2} = 1 = \frac{d}{d}, \quad (17)$$

$$\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = |\mathcal{O}_1 \cup \mathcal{O}_2| \cdot \frac{d-1}{d} = 12 \cdot \frac{3}{4} = 9 = (d-1)^2, \quad (18)$$

$$\text{Tr}(P_{\text{ker}}) = 1 + (d-1)^2 = 10 = \dim \ker(H_{\text{wind}}). \quad (19)$$

4.2 The EFT coefficient c

Theorem 5 (EFT coefficient from the kernel diagonal). *The EFT coefficient of HCSM-52 satisfies*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{d+1}{(d-1)^2} = \frac{5}{9}. \quad (20)$$

Proof. Numerator: $d+1 = 5$ (framework constant). Denominator: the partial trace of P_{ker} on the 12 modes of $\mathcal{O}_1 \cup \mathcal{O}_2$, computed above as $(d-1)^2 = 9$. Ratio: $5/9$. $\square$

4.3 The mass-map coefficient

Theorem 6 (Mass-map coefficient from the kernel diagonal). *The mass-map coefficient of HCSM-19 and HCSM-33 satisfies*

$$\frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{(d-1)/d}{2/d} = \frac{d-1}{2} = \frac{3}{2}. \quad (21)$$

Proof. Ratio of the two values of the diagonal of P_{ker} : $\frac{3/4}{1/2} = 3/2 = (d-1)/2$. $\square$

4.4 The kernel dimension

The kernel dimension $\dim \ker(H_{\text{wind}}) = 10$ is the total trace of P_{ker} , decomposing as $1 + (d-1)^2$ over the two orbit classes. This reproduces the constant used in HCSM-53's dark-energy formula (the $|F|$ denominator appears as $\dim \ker$ in the alternative derivation).

4.5 Unification

The three coefficients $c = 5/9$, $(d-1)/2 = 3/2$, and $(d-1)^2 + 1 = 10$ are three traces of the same object: the diagonal of the projector onto $\ker(H_{\text{wind}})$. They are not independent fits; they are the same structural fact read on three different orbit unions.

5 F-Invariance of the Kernel

Theorem 7 (F -invariance). *The kernel of H_{wind} is invariant under the diagonal flip F , but not under the full D_4 point group:*

$$[P_{\text{ker}}, P_F] = 0, \quad [P_{\text{ker}}, R(r)] \neq 0, \quad [P_{\text{ker}}, R(s)] \neq 0. \quad (22)$$

At machine precision, $\|[P_{\text{ker}}, P_F]\|_F < 10^{-14}$ while $\|[P_{\text{ker}}, R(r)]\|_F = \|[P_{\text{ker}}, R(s)]\|_F = \sqrt{2}$.

Proof. By HCSM-14 Theorem 20, $\{H_{\text{wind}}, P_F\} = 0$. Squaring, $H_{\text{wind}}^2 P_F = P_F H_{\text{wind}}^2$, so P_F commutes with the spectral projector of H_{wind}^2 at eigenvalue 0. Since $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$, this gives $[P_{\ker}, P_F] = 0$.

For the nonexistence with r and s : the generator $R(r)$ of order 4 does not preserve the F -cycle structure. Direct numerical evaluation of $[P_{\ker}, R(r)]$ gives Frobenius norm $\sqrt{2}$; the same for $[P_{\ker}, R(s)]$. $\square$

Corollary 8. *The kernel of H_{wind} is the unique F -invariant 10-dimensional subspace of $\mathbb{C}^{14}$ that contains the class $\{\text{dark}, W, Z, H\}$ up to the F -swap.*

6 The Kernel- T Decomposition

Theorem 9 (Kernel splitting under the processor). *Let $T = I - P_C$ be the processor of HCSM-13, with image F (rank 10) and kernel $C = \{\text{dark}, W, Z, H\}$ (rank 4). Then*

$$\dim(\ker(H_{\text{wind}}) \cap \text{im}(T)) = 6 = |F| - |B|, \quad \dim(\ker(H_{\text{wind}}) \cap \ker(T)) = 4 = |B|. \quad (23)$$

The two dimensions sum to $10 = \dim \ker(H_{\text{wind}})$.

Proof. Numerical. Form the stacked matrix $[U_{\ker} \mid U_F]$ with U_F an orthonormal basis of $\text{im}(T)$. The rank is 14 and the nullity of the coefficient system is 6; equivalently $\dim(\ker H_{\text{wind}} \cap \text{im } T) = 6$. By rank-nullity the complementary intersection has dimension 4. $\square$

Remark 10. The decomposition $\ker(H_{\text{wind}}) = (6\text{-dim}) \oplus (4\text{-dim})$ under the processor T mirrors the F/B split of HCSM-13. The kernel of the winding generator is not the same as the kernel of the processor; it is a 10-dimensional subspace that intersects the bosonic sector B in 4 dimensions and the fermionic sector F in 6 dimensions.

7 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-60_verification.py` constructs:

1. The 14-mode coordinates and the winding matrix M .
2. The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$.
3. The kernel basis $U_{\ker}$ by SVD.
4. The projector $P_{\ker} = U_{\ker} U_{\ker}^T$.
5. The permutation matrices $P_F, R(r), R(s)$.

and verifies:

1. $(P_{\ker})_{ii} = 1/2$ for $i \in \mathcal{O}_0$.
2. $(P_{\ker})_{ii} = 3/4$ for $i \in \mathcal{O}_1 \cup \mathcal{O}_2$.
3. $\text{Tr}(P_{\ker}|_{\mathcal{O}_0}) = 1$.
4. $\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 9$.

5. $\text{Tr}(P_{\text{ker}}) = 10$.
6. $c = 5/9$.
7. $(d - 1)/2 = 3/2$.
8. $\| [P_{\text{ker}}, P_F] \|_F < 10^{-14}$.
9. $\| [P_{\text{ker}}, R(r)] \|_F = \sqrt{2}$, $\| [P_{\text{ker}}, R(s)] \|_F = \sqrt{2}$.
10. $\dim(\ker(H_{\text{wind}}) \cap \text{im } T) = 6$.
11. $\dim(\ker(H_{\text{wind}}) \cap \ker T) = 4$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

8 Conclusion

The kernel of the winding generator carries the framework's three most consequential coefficients on its diagonal. The EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ are traces of one projector, read on three orbit unions.

The single structural fact behind all three is the diagonal flip F . By HCSM-14, H_{wind} anti-commutes with F ; therefore its kernel is F -invariant; therefore the diagonal of the kernel projector takes one value on each F -cycle. Each F -cycle sits inside a single D_4 orbit, and the two orbits $\mathcal{O}_1$ and $\mathcal{O}_2$ happen to carry the same value.

The pattern $\{2/d, (d - 1)/d\}$ is the fingerprint of the winding kernel. It is not a numerical accident; it is a theorem of the substrate. One projector, two values, three coefficients.

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